Machine Learning Project

Cardiac Arrest Prediction System

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Abstract

Early cardiac arrest in the emergency department (ED) poses a significant risk to patient survival, as timely intervention is critical. This study proposes a real-time Early Cardiac Arrest Alert system that continuously monitors patients' vital signs—including heart rate, blood pressure, oxygen saturation, respiratory rate, and body temperature—using machine learning. The system employs a Long Short-Term Memory (LSTM) model to detect temporal patterns in physiological data and predict potential cardiac arrest events 10 to 30 minutes in advance. The methodology involves preprocessing the data to handle missing values and noise and training the LSTM model to identify early risk indicators. Results show that the system outperforms traditional approaches in predicting accuracy and response time, providing actionable alerts to medical staff. Additionally, the system highlights key contributing risk indicators, enabling targeted intervention. In conclusion, integrating machine learning with continuous patient monitoring in the ED significantly enhances early cardiac arrest detection, reduces mortality rates, and improves emergency care efficiency.

Keywords:

• **Machine learning:** Techniques that allow computers to learn patterns from data and make predictions.

• **LSTM:** A neural network model that handles time-series data by remembering long-term patterns.

• **Cardiac arrest prediction:** Estimating the risk of a patient experiencing a cardiac arrest.

• **Real-time monitoring**: Continuously tracking patient data as happens.

• **Emergency department**: The hospital unit that treats urgent and critical cases.

• **Vital signs analysis:** Examining key body measurements like heart rate and blood pressure to assess health.

Introduction

Cardiac arrest in emergency departments remains a leading cause of sudden patient death worldwide. Rapid detection and intervention are crucial, yet current monitoring systems often fail to provide early warnings due to their reliance on threshold-based alarms or single vital sign measurements. Machine learning offers a promising approach to overcome these limitations by learning complex patterns from multi-modal, time-series of physiological data. This review focuses on designing a real-time predictive system for early cardiac arrest detection using LSTM networks. The objectives are to explore how multi-vital sign analysis can improve predictive accuracy, reduce response times, and enhance patient outcomes in the ED. The scope includes reviewing current predictive approaches, discussing methodological frameworks, and proposing improvements for real-time, data-driven alerts.

Problem Definition

The chosen problem is the significant delay in detecting cardiac arrest in the emergency department, where traditional methods often take 3–4 hours to identify high-risk cases, increasing the likelihood of patient mortality due to late intervention. Current systems typically rely on fixed thresholds or single vital sign monitoring, limiting their predictive capability in real-time. The motivation for improvement lies in leveraging machine learning to analyze continuous, multi-modal vital sign data and reduce the prediction time to 10–30 minutes before the cardiac arrest occurs, providing the medical staff with a critical early intervention window. Additionally, the system will explain the reason behind each prediction by highlighting the contributing vital sign indicators, enabling medical personnel to understand the risk and take immediate action. The suitable approach involves using an LSTM model designed for time-series data, capturing temporal dependencies, and generating accurate real-time alerts, ultimately improving patient safety and enhancing decision-making in critical care environments.

Literature Review

A. Current State of the Art

Paper / Model	Data Source, Size	Features	Model / Techniques	Key Performance	Key Innovations
1. Prediction of Cardiac Arrest in ED (Samsung, Korea)	214,307 ED visits, 993 cardiac arrest events.	Static: age, sex, chief complaint; Sequential: 6 vital signs	Logistic Regression (LR), Random Forest (RF), LSTM-RNN	Best RF: AUROC 0.97, AUPRC 0.86, Sens 0.94, Spec 0.90	vital sign modeling, class imbalance handling.
2. Nationwide ML Algorithm (South Korea)	1,350,693 patients, 5,431 IHCA events.	Patient-level: demographics, vitals, labs; Hospital-level: ED crowding, beds	LR, XGBoost, MLP	Best XGBoost: AUROC 0.927, AUPRC 0.132	Combined pre-hospital, ED data, SHAP interpretability.
3. Deep EDICAS (NTUH, Taiwan, Yen)	120,000+ ED patients, 250 cardiac arrests.	Vital signs, demographics, triage	LSTM + Attention	LSTM: AUROC 0.93, AUPRC 0.51,	Data augmentation for class imbalance, integrated multivariate time-series encoding, early CPR prediction
4. Smart Cardiac Framework ML Algorithm (SCF)	18 participants 540 real-time sample.	vital signs, basic demographics	ANN, Random Forest, XGBoost, SVM, Naïve Bayes, Decision Tree	Best Decision Tree: AUROC, AUPRC around 96–98%	Low-cost continuous monitoring design Real- vital-sign

Table 1: State-of-the-Art Techniques in Cardiac Arrest Prediction

Insights from Current State of the Art:

Recent studies on predicting sudden cardiac events in the Emergency Department (ED) highlight several key trends and challenges:

• Sequential/time-series modeling (e.g., LSTM, Random Forest) is crucial for capturing dynamic changes in patients' vital signs and accurately predicting cardiac events.

• Handling class imbalance remains a major challenge due to the low incidence of cardiac arrest events.

• Interpretability techniques, such as SHAP values and Attention mechanisms, are increasingly incorporated to enhance clinical trust and facilitate decision-making.

• Data limitations exist, as most studies rely on single-center datasets, with external validation and multi-hospital generalizability still limited.

Overall, these insights emphasize the importance of advanced sequential modeling, robust strategies for early cardiac arrest prediction, and model interpretability, while highlighting the need for broader validation across multiple centers.

B. Comparative Analysis

Aspect	Decision Tree	Random Forest	XGBoost	LSTM / Deep EDICAS
Strengths	Simple, fast, interpretable	Captures non-linearities, robust	High predictive accuracy, handles large feature sets	Captures temporal dependencies, dynamic, static data integration
Weaknesses	Overfitting, limited with high variability sensor data	Less interpretable, may overfit	Computationally intensive, precision limited in rare events	Computationally intensive, require large data
Best Use Case	Small datasets, fast real-time alerts	Small-to-medium datasets with sequential features	Large tabular datasets for prediction tasks	Continuous monitoring, real-time prediction with temporal data

Table 2: Comparative Analysis of Machine Learning Models for Cardiac Arrest Prediction

Observations:

• RF and XGBoost perform well with structured tabular data; LSTM excels when temporal sequences are critical, decision tree for fast early prediction.

• Attention mechanisms enhance interpretability without sacrificing predictive power.

• Short-term prediction (minutes vs hours) is an active research focus to increase clinical utility.

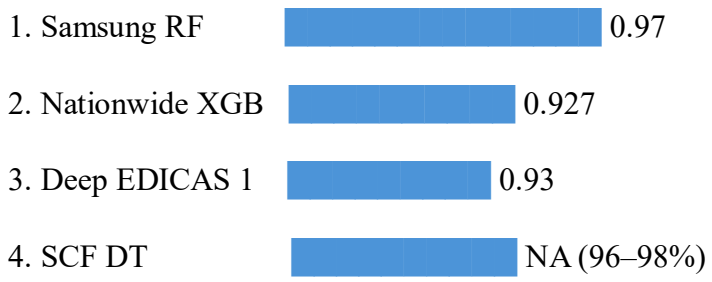
Gap Analysis and Justification

Gap / Limitation	Evidence	Justification for Our Project
Single-center bias	Deep EDICAS, Samsung study	Need multi-hospital data to improve generalizability
Class imbalance	Rare cardiac arrest events, low AUPRC in nationwide study	Use advanced oversampling / augmentation and short-window predictions to increase sensitivity
Limited real-time application	Nationwide study and Deep EDICAS 1–3h windows	Implement 10–30 min streaming predictions for immediate clinical alerts
Incomplete feature set	Samsung, nationwide, SCF studies exclude some labs, medication data	Include additional vitals, lab trends, and patient history for more accurate predictions
No temporal modeling	Samsung, nationwide, SCF used models (DT, RF, NB, ANN) rely on static samples	Implement sequential models (LSTM/GRU/Transformers) to capture minute-by-minute physiological changes

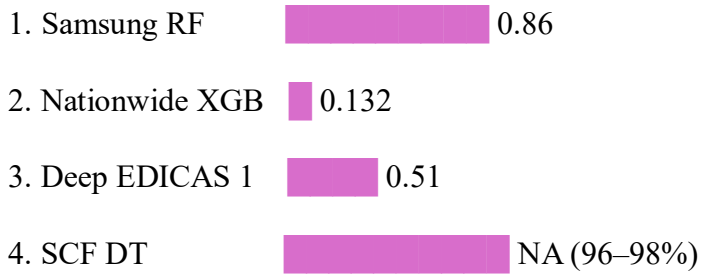
Table 3; Gap Analysis and Justification for Early Cardiac Arrest Prediction

Visual Comparison (Conceptual)

- AUROC Performance (Higher is better)



- AUPRC Performance (Handles rare events)



Interpretation:

• RF excels with balanced datasets for high AUROC and AUPRC.

• Nationwide XGBoost shows high AUROC but struggles with rare event precision (AUPRC low).

• Deep EDICAS balances temporal modeling and interpretability, achieving strong AUROC and moderate AUPRC.

• SCF (2021) did not report exact AUROC/AUPRC values, but mentioned that the Decision Tree achieved an approximate AUROC of 96–98%

Detailed ML Module

Objectives

The ML module aims to:

- Detect high-risk patients in the emergency department **10–30 minutes before a potential cardiac arrest**.
- Provide **explainable predictions** by highlighting the most influential vital signs contributing to each risk assessment.
- Generate **interpretable alerts** to support real-time decision making by medical staff.
- Reduce patient mortality by enabling **early intervention** through continuous monitoring and predictive analysis.

Approach

The proposed module uses a **Long Short-Term Memory (LSTM) neural network**, a model designed for sequential and time-series clinical data.

Why LSTM?

- Suitable for **time-series data** and can capture temporal dependencies in vital signs.
- Dataset contains multiple measurements per patient (20 records per patient), which allows the model to learn from **sliding-window sequences** of length T=5.
- **Predict cardiac arrest 10–30 minutes in advance**, providing an early intervention window.

Enhancements:

- **Lookahead mechanism:** Predicts cardiac arrest several minutes before occurrence.
- **Class weighting:** Handles **imbalanced datasets** with more low risk than high risk patients, improving model fairness.
- **SHAP values:** Shows feature importance for each prediction, improving **interpretability**.
- **Early Stopping:** To prevent **overfitting** and enhance generalization.

Features:

The ML module processes the following real-time patient vital signs and clinical indicators:

- Heart Rate
- Respiratory Rate
- Body Temperature
- Oxygen Saturation
- Systolic / Diastolic Blood Pressure
- Age, Gender, BMI
- Derived MAP, HRV, Pulse Pressure

Output:

The ML module produces:

- Predicted **risk category** (High Risk / Low Risk).
- **Prediction probability** for cardiac arrest.
- **Explainable indicators** highlighting which features contributed most to the prediction.
- **Real-time alerts** for abnormal vital signs.

Architecture / Workflow Diagram

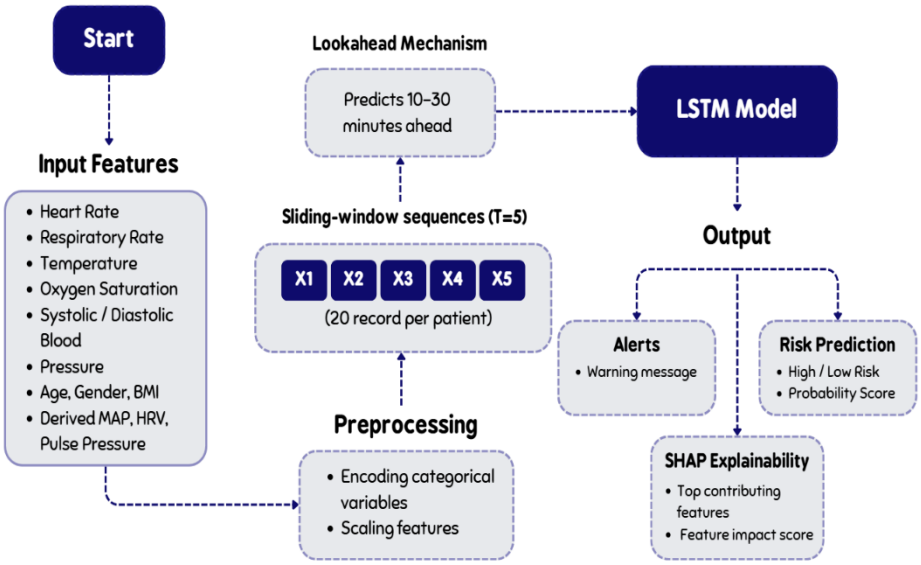


Figure 1: LSTM Model Workflow for Cardiac Arrest Prediction

The architecture of the LSTM model in Figure 1 follows a structured data path to deliver early and interpretable predictions:

1. Input Features:

- The system continuously receives real-time patient data**.**

2. Preprocessing and Sequencing:

- Categorical variables encoded and numerical features scaled to ensure optimal model performance.
- Then data grouped into sliding-window sequences of length T = 5, representing continuous monitoring periods that the LSTM model uses for time-series analysis.

3. LSTM Model and Lookahead Mechanism:

- The sequences are passed through **a** Long Short-Term Memory (LSTM) network**,** chosen for its ability to capture long-term dependencies in time-series vital signs.
- A Lookahead Mechanism enables the model to generate predictions 10–30 minutes in advance, providing a critical early intervention window for clinical staff.

4. Outputs and Interpretability:

The system generates three key outputs to support clinical decision-making:

1. **Risk Prediction:** Categorizes patients as High Risk or Low Risk, along with a probability score.
2. **Alerts:** Sends immediate warning messages for critical vital sign changes.
3. **SHAP Explainability:** Highlights the most influential variables driving each prediction, ensuring transparency, and enhancing clinicians' trust in the system.

Integration

The ML module integrates into the hospital monitoring system:

- Receives real-time patient vital signs from medical devices.
- Processes the data continuously and generates risk predictions.
- Sends alerts to medical staff **before cardiac arrest occurs**.
- Provides SHAP-based explanations to support **clinical decision making**.

Expected Benefits

- **Early Detection:** Reduces the time to detect cardiac arrest from hours to 10-30 minutes.
- **Actionable Insights:** Alerts staff about critical vital signs, enabling timely intervention.
- **Improved Accuracy:** LSTM captures temporal trends, leading to more reliable predictions.
- **Interpretability:** SHAP explanations help staff understand the “why” behind predictions.
- **Enhanced Patient Safety:** Real-time monitoring and prediction improve patient outcomes and reduce mortality risk.

Results / Model Performance

Confusion Matrix

The confusion matrix as shown in Figure 2 illustrates the model's predictions against the true labels. The LSTM model correctly identifies most high-risk patients with very few misclassifications.

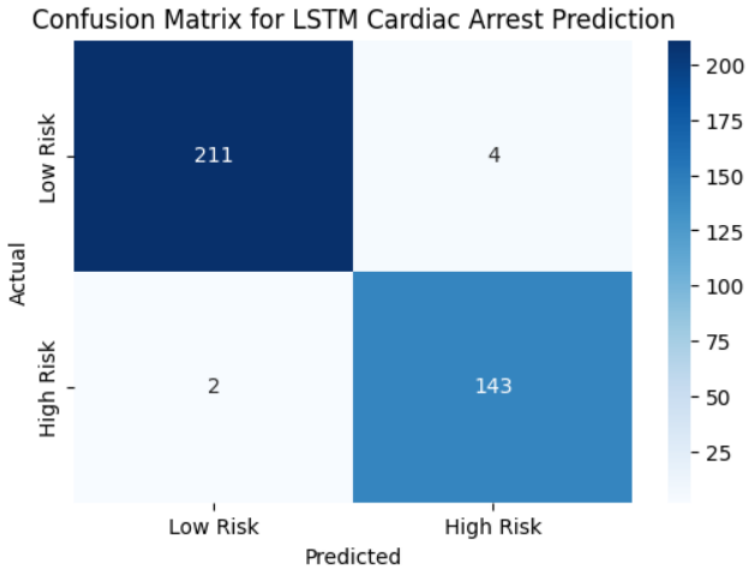


Figure 2: Confusion matrix of the LSTM model.

Performance by Class

Class	Low Risk	High Risk
Precision	0.99	0.97
Recall	0.98	0.99
F1-score	0.99	0.98

Table 4: Classification Performance of LSTM Model.

Overall Model Performance

- Accuracy: 0.983 → 98.3%
- ROC AUC: 0.999 → perfect in distinguishing between the two classes as shown in Figure 3.

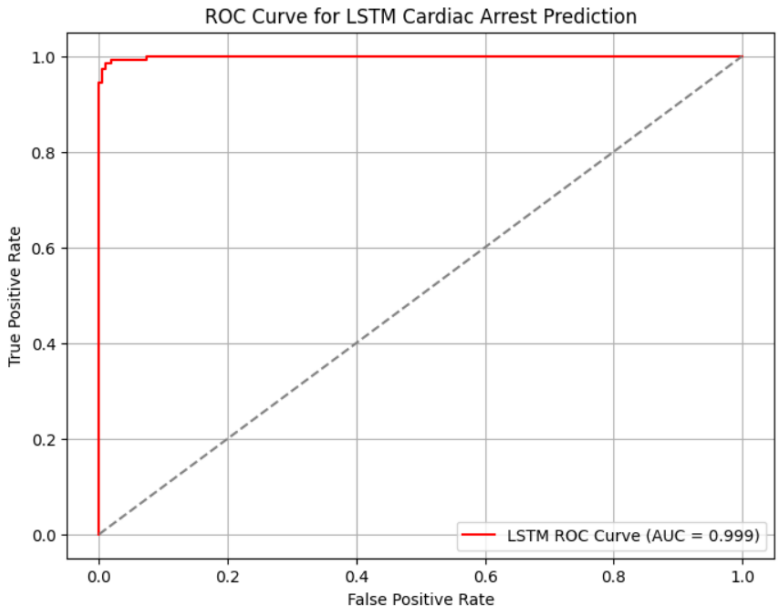


Figure 3: ROC Curve for LSTM model.

Interpretation:

- The model achieved **high accuracy (98%)**, correctly predicting most Low Risk and High Risk patients.
- **Precision and recall** are balanced, indicating strong performance for both classes.
- **ROC AUC = 0.999** shows excellent discrimination capability.
- Confusion Matrix highlights only **5 misclassifications** out of 360 samples, demonstrating reliable early risk detection.

Training and Validation Curves

As shown in Figure 4 and Figure 5 training and validation curves show that the model converges without overfitting. Loss decreases smoothly while accuracy rises steadily, confirming stable training.

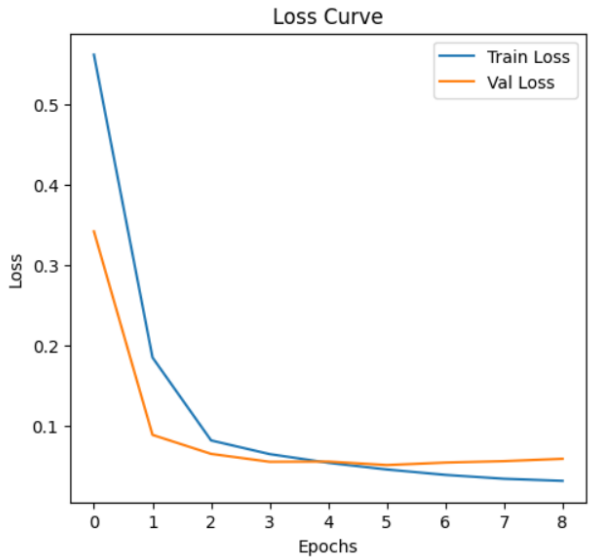


Figure 4: Training vs. Validation Loss curve.

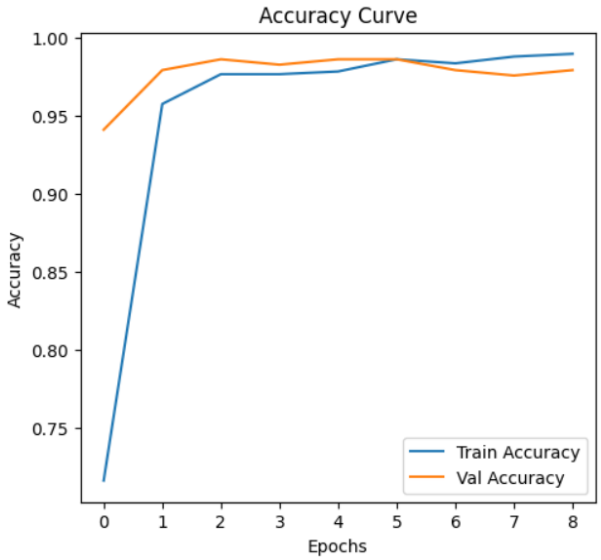


Figure 5: Training vs. Validation Accuracy curve.

Example Output

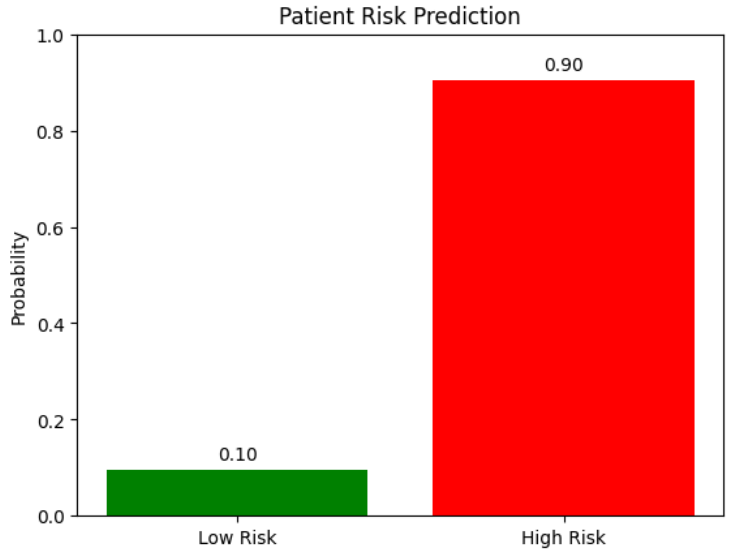
This section presents sample outputs from the Cardiac Arrest Prediction system as shown in Figure 6.

```
Enter Patient ID (or 0 to exit): 7

🧍 Patient ID: 7
✓ Actual Risk: High Risk
✓ Predicted Risk: High Risk 0.90

🚨 Alerts:
⚠️ Heart Rate abnormal (101.57)
⚠️ Respiratory Rate abnormal (26.43)
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step
100% ━━━━━━━━━━━━━━━━━━━━ 1/1 [00:20<00:00, 20.98s/it]
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 52ms/step
6775/6775 ━━━━━━━━━━━━━━━━━━━━ 13s 2ms/step

🔍 Top 5 Factors Influencing Prediction:
- Oxygen Saturation: -0.39 (Low Risk)
- Heart Rate: 0.25 (High Risk)
- Systolic Blood Pressure: 0.16 (High Risk)
- Derived_MAP: 0.12 (High Risk)
- Respiratory Rate: 0.00 (Low Risk)
```




```
Enter Patient ID (or 0 to exit): 0
Program terminated.
```

Figure 4: Example of Cardiac Arrest Prediction Output.

This output illustrates how the system provides real-time risk prediction, alerts for abnormal vital signs, and interpretable feature contributions to support clinical decision making.

Limitations of the ML Module

- **Limited Data Availability:** The available dataset was mostly restricted or incomplete, which forced the team to generate synthetic or simulated data to train and test the model.
- **Potential Overfitting / Unrealistically High Performance:** The extremely high accuracy and ROC AUC scores suggest that the model may have overfitted the limited or synthetic data, reducing its generalizability to real-world scenarios.
- **Dependency on Continuous and Accurate Measurements:** The model's predictions rely on real-time, precise vital signs. Any missing or noisy data could degrade performance.
- **Single-Center or Non-Diverse Data:** Due to the limited dataset sources, the model may not generalize well across different hospital settings or patient populations.
- **Possibility of False Alerts:** While the system provides real-time warnings, false positives or false negatives may occur, potentially affecting clinical decision-making.

Conclusion

This study presented a real-time early cardiac arrest alert system in the emergency department using an LSTM-based machine learning model to analyze patients' time-series vital signs. The system demonstrated very high performance, with an accuracy of 98.3% and a ROC AUC close to 1.0. However, these results may be overly optimistic due to the limited availability of real data and partial reliance on synthetic data, indicating potential overfitting. The system provides 10–30-minute advance predictions, offering a critical early intervention window and actionable alerts for medical staff.

The findings highlight the need for future research to enhance model reliability and generalizability, including:

- Using large-scale, multi-hospital real-world datasets to validate the model.
- Incorporating additional data sources, such as lab tests, medications, and patient history, to improve prediction accuracy.
- Developing multimodal models that combine medical imaging, vital signs, and electronic health records.
- Enhancing interpretability techniques to strengthen clinical trust and enable faster decision-making.
- Investigating methods to reduce false alerts and improve real-time alert quality, minimizing alarm fatigue for medical personnel.

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